Engineering Mechanics 1 · Unit_02 · UNIT 2

# DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 17 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents

1. 1 · UE26ME142A Unit 2 Class 17 AVSummary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

# UE26ME142A Unit 2 Class 17 AVSummary

## 1 UE26ME142A Unit 2 Class 17 AVSummary

Study focus: UE26ME142A_Unit 2_Class 17_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

My name is Dr.

**Key details**

- Svink Tesh from Civil Engineering under the Department of Science and Humanities PS University.
- This is class number 21.
- problems on equations of equilibrium.
- Which states that the ATKG exerciser is beginning to execute some slow, steady bicep currents.
- As tension T is equal to 65 newton is developed against an exercise machine not shown.
- Determine the normal reaction forces.
- feet A and B friction is sufficient to prevent slipping and the exerciser maintains the position shown with the center of mass at G.
- Under the feet A and B, and he also says that the friction is sufficient to prevent slippage.
- There's a tension T, there's a machine here which is not indicated.
- The direction of this tension T with respect to the horizontal is 20 degrees.
- The centre of mass of the person.
- is G where the 80 kg mass is going to be acting and of course if you convert it into a weight it will be 80 into 9.81 newtons which will be acting at G.
- Then you have two points are the under the feet of the person.
- Point A and point B, you will have a normal reaction exerted at A, the normal reaction exerted at B and there will be a friction force.
- One is the tensile force acting to the right of the screen, which is 65 newtons.
- Then the sulphate of the person which is 80 kgs into 9.81, which will be acting at point G.

- You will have a normal reaction N acting upwards at A, normal reaction at B acting upwards, frictional force at A which is acting to the left of the screen, frictional force at B acting to the left of the screen.
- All the forces that I have discussed is indicated here along with the dimensions.
- Then the sign convention is also indicated here.
- X is positive, y is positive up third, x is positive to the right.
- Then movement, anti-clockwise movement is taken as positive.
- The horizontal component of the tension will be equal to the frictional force that is going to be acting F A and F B, but we are not interested in that.
- What we are interested is in N A and N B.
- There are two unknowns.
- The horizontal summation of all the forces along horizontal cannot be taken, so we can take summation of all the forces along vertical algebraic sum.
- Then we can take movement about any point to be equal to zero.
- So first equation summation of all the forces acting along Y to be equal to zero.
- Plus or this is negative, NEA is positive, N B is also positive, this is negative, then the component of this T will be acting upwards, so it is positive.
- That is 65 sin 20.
- That should be equal to 0 and that is what is indicated here.

**Example or application from the lecture**

- look at what is given in the statement.
- They have given you the mass of the person, which is 80 kgs.
- They have also given you what is the tension in the cable, which is T that is equal to 65 newtons, and what is asked is to determine what is the normal reaction.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

Next taking moment about any point, it is better to take the moment at the point where you have the support reactions.

**Key details**

- We get NA.
- into the distance between A and B which is 500 millimeters.
- Then you have this sulfate of the person is 18 to 9.81 into the horizontal distance between G and B is 250.
- Then we have.
- The component of T along the horizontal, which is 65 into cos 20 into the vertical distance, that's what is indicated here.
- Then we have a component of T which is acting along the vertical, which is 65 sin 20, into the distance from B to C, which is 350 mm, and that's what is indicated here.
- We will get NA equals 219 from this.

- Substituting the value of NA here, we get the value of N B which is equal to 544 m.
- I hope you have understood here.
- Of course the algebraic sum of all the forces along the horizontal if you take, you will have FA plus F B to be equal to 65 cos 20 degrees.
- I hope all of you have understood this.
- The statement of the problem is determine the moment which the motor e must exert.
- In order to position the uniform slender bar of mass m and length L in the arbitrary position theta, the ratio of the radius of the gear whe attached to the bar.
- To that of the gear wheel A, attach it to the motor shaft is 2.
- What am I supposed to apply the movement here such that this bar of length L would be in the position in an arbitrary position.
- That is at an angle theta is what is asked.
- If this has to rotate in the anti-clockwise direction, then this arm will also rotate along with that and it will come to the or in this arbitrary position at an angle T.
- If this has to rotate in this way,.
- Then the movement that needs to be applied here, so here will be in the clockwise direction.
- When I apply a movement here, so when I apply this movement, you will have a force that is going to be acting along this tangent line.
- The weight of this would be acting at the center.

**Example or application from the lecture**

- No, if there is no movement say consider at this point, then this arm would simply rotate in the clockwise direction and this will come to a vertical position.
- No, they have also given you the ratio of the radius of the shaft that is this and this or in the ratio of 2, or rather if I take this as R, then this would be 2 R.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

The sulfate of the of this the O C would be acting vertically down which is equal to Mg.

**Key details**

- The force that is acting on the gear would be like this, which is F, because this would produce a clockwise movement to prevent it from rotation, I should apply a force F here, which will act in the anti-clockwise direction.
- sigma m or b is equal to 0, f into 2 r minus mg into this horizontal distance would be this is L by 2 sine of theta that's what is indicated here.
- M, G, and L as well as T.
- Now this force F which is acting here would be generated because of the gear system at A.
- Here, the fourth that is going to be acting here will be in the opposite direction.
- Whereas this force would act in the anti-clockwise direction.
- We will get this as R and R will get cancelled.

- That should be the moment that should be produced in the motor that is present at the point A, that is this one.
- You have the motor, the movement that should be produced in this motor, which should rotate in the clockwise direction, should be equal to one fourth of your MGL sin theta.
- Then it is also depends upon at what angle this bar is to be rotated.
- I hope you have understood.
- The equations of equilibrium that is summation of the forces along X, algebraic sum of all the forces along X, algebraic sum of all the forces along Y, then movement about any point should be equal to 0.
- But before using these three equations,.
- I am able to draw a clear free body diagram, right?
- Isolating what is that I want.
- As you can see here, I have taken the pulley at B and the rod that is acting at B separately along with the forces that is acting on this body.
- That is the force of.
- The sulfate of the rod okay is what I've taken.
- What are this B will now act as an hinge.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

**Core explanation · 15:00–19:59**

Okay, so this also you can see that it is like an hinge, and because of that, you have these forces a y and ax.

**Key details**

- Of course, nowhere I have taken.
- By, Bx as well as AY and AX, because in both the cases what I have considered is summation of all the movement about B is equal to 0, summation of all the movements about A is equal to 0.
- I have not used the vertical force, algebraic sum of vertical force, or algebraic sum of the horizontal forces because that is not what I intend to determine.
- They have not asked us to determine Bx or BY as well as Ax or AY.
- If they would have asked, I can easily use summation of all the forces along vertical to be 0, summation of all the forces along horizontal to be equal to 0, and then I could have determined what is Bx and BY.
- Since I have got F here, okay, this F.
- And I can also determine what is the angle made by this force with horizontal or vertical.
- I can resolve this along vertical and horizontal, and I will get what is Bx in terms of F and also BY in terms of F and the sulfate of the rod.
- But in the problem, since they have not asked us to determine the reactions acting at B or A, we are not finding out that.
- A statics by Miriam and Craig and Bolton published by Wiley Publication and it is 8th edition.

- Indian edition itself.
- We'll be continuing solving of few more problems in the next class.
- My name is Dr.
- Sri Vengtesh from Department of Civil Engineering under Department of Science and Humanities, PES University.
- We will meet in the next class.

**Example or application from the lecture**

- Okay, and all the problems that are solved are example problems from the prescribed textbook, that is engineering mechanics.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| UE26ME142A_Unit 2_Class 17_AVSummary transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| UE26ME142A_Unit 2_Class 17_AVSummary transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| UE26ME142A_Unit 2_Class 17_AVSummary transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps. |
| UE26ME142A_Unit 2_Class 17_AVSummary transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

## Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.

9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00014 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare... |
| SRC00015 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00016 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare... |
| SRC00017 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare... |

# C Revision and retrieval practice